Chapter 1

Course Overview and Short Intro to R

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MGMTMSA 400

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Introduction

Introduction to Your Instructor

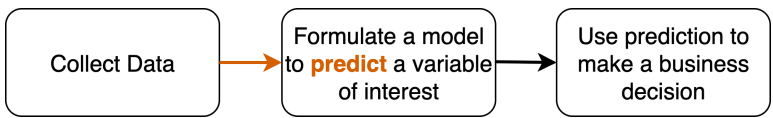
- Ph.D. in Marketing, The Wharton School (University of Pennsylvania)
- Joined UCLA Anderson as an assistant professor in July 2025
- Research area: Empirical industrial organization, with a focus on bargaining and the video streaming market

Syllabus

- **Software:** You may use any software for assignments, but only R will be taught in lectures and TA sessions
- **Assignments:** Group submissions via **Gradescope**. **No late work accepted.**
- **Exams:**
  - Scheduled for 10/27 and 12/2
  - Closed-book, multiple choice only
  - Scientific calculator and one double-sided page of notes permitted
  - Exams from previous years will be shared in advance
- **Attendance:** iClicker check-in required in Lectures 3–7
- **Participation:** Meaningful contribution in class is strongly encouraged

Introduction to Data Analytics

- What is Data Analytics?
- Data analytics has three key elements
  - Manipulate and transform data
  - Build and estimate models that can be used to predict consumer and firm behaviors
    - "Behavior" could be market valuation, product sales, response to TV content
  - Methods to evaluate, compare and select models



Netflix's Price and Content Choice: A Data Analytics Example

- **Problem:** Starting in 2026, Hulu and Disney+ will merge into a single platform. Netflix wants to **optimize** both (a) subscription price and (b) content portfolio to respond effectively
- **Challenge**: Such adjustments are costly and time-consuming to implement
- **Solution**
  - Modeling: Develop a model that relate subscription prices and content libraries to consumers' subscription choices across all streaming services, using **historical** data
  - Prediction: Formulate a profit-maximization problem (non-linear programming) with constraints such as budget limits and competition

The Value of Models

- Optimization is at the heart of many business decisions
- Any attempt at optimization must involve predicting a counter-factual
  - E.g., how many subscribers would Netflix lose if it removes *Wednesday* from the library?
- A model is essential to answer this question
  - Structure historical data
  - Provide the benefit of interpolation in high dimensions

Scope of the Course

Big Data

There are two types of Big Data:

- **Too Large**
  - Data that is so large that manipulating the data for simple summaries like means or tables is a challenge even in a relatively powerful computing environment
  - Example: Google search data: 100 billion search queries a month; 30 trillion URLs
  - This is a computer science problem not an analytics problem
- **Too Complicated**
  - Data that is very high dimensional. There are a lot of variables that may affect the variable of interest
  - Example: Product recommendation system data: Too many consumers and too many products
  - This is a problem which requires structure and model building based on that structure
- We will focus on the second type

Machine Learning: Unsupervised vs. Supervised

There are two types of Machine or Statistical Learning:

- **Supervised Learning**
- **Unsupervised Learning**
- We will focus on supervised learning

Supervised Learning

- There are a **target** or **output** variable (y) and a set of **input** variables (X)
- The goal of supervised learning is mostly **predictive**: Given a set of X, what is the best predicted value of y?
- Two examples of supervised learning
  - **Classification**: We have a set of consumer loans and we observe whether or not the consumer defaults on the loan. The input variables are characteristics of the borrower that we obtain from application data
  - **Regression**: We have sales as the output variable and the input variables are the characteristics of the product and competing products (e.g. prices and product features)
- Our course will focus on supervised learning!
  - We will focus on what are called "classical" methods of ML. You will get an exposure to "deep" learning in Professor Maoz's class

Unsupervised Learning

- There is no **target** variable (y); only a set of **input** variables (X)
- The goal of unsupervised learning is to obtain some insights of the "structure" of X. No y is used or observed in this process
  - The "structure" is usually a lower dimensional summary of X
  - We call these summaries "features" or "embeddings"
- An example of supervised learning
  - We might have ratings on 100 aspects of a service experience such as a hotel visit (overall satisfaction, price, cleanliness, comfort of the bed, room service, ...)
  - Each one of these dimensions of the hotel product experience is summarized by a quality rating (say 1-7 points from "extremely dissatisfied" to "extremely satisfied")
  - The goal of a **unsupervised** learning task would be to explore the inter-relationships between these quality ratings to understand what **lower dimensional** structure exists
  - For example, we might find that these ratings can be reduced into three dimensions: value, comfort, and amenities
- This is not the focus of our course!

Causal vs. Correlational Effect

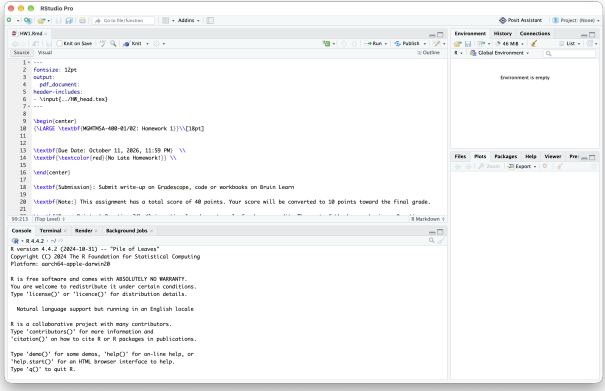
- **Correlational Effect**
  - Captures an *association* between two variables
  - Example: MacBook has a higher price and higher sales compared to Dell Inspiron $\implies$ higher prices are *associated* with higher demand
  - But this is *not causal*: a higher price should not lead to higher demand!
- **Causal Effect**
  - Measures the effect of changing one variable on another, *holding all else constant*
  - The MacBook vs. Dell example is only correlational, not causal
- Be very cautious when interpreting correlations as causality

A Short Intro to R

R, R-Studio and course package, DataAnalytics

- R is a free package with versions for Windows, MAC OS, and Linux. Visit CRAN site nearest you (see above or google "CRAN") and download and install R
  - R is one of the standard languages of data science
  - R is a powerful and extensible language. More than 10,000 extension packages have been developed. Freeware is a virtue not a negative!
- R-Studio is a programming environment for R. After installing R, download and install R-Studio
- **Warning**: We provide python code snippets in parallel with R for those who want to translate the R concepts into python. However, we will not display or discuss Python codes in class

The R Studio Environment



Four panels in Rstudio:

- Top Left: R script
- Bottom Left: Console
- Top Right: Workspace, History
- Bottom Right: Files, Plots, Packages, Help

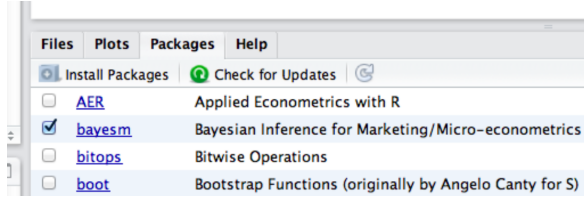
R Scripts

- Download `Ch 1 code snippets.R` file from the BruinLearn site and "open" the file. The contents will appear in the upper left hand corner of RStudio
  - This is an "R script" or collection of R commands
- There are two ways to execute an R command:
  - Type in the Console window (lower left)
  - Highlight a line in a script (if open) and click 

- **Suggestions**
  - It is best to type commands into the script window and execute so that you can correct and iterate
  - Save the script file in some logical place on your computer and you can return to your commands
  - Advantages: You can replicate your analysis!

R packages

- R is designed to be extendible. When you install R, you are installing a very large set of commands and datasets that are called "base R"
- To add capabilities and data, you need to install what are called "packages." Packages are bundles of code, datasets, and documentation
  - Most packages can be found on the world-wide network of mirror sites called CRAN. But others can be found on BitBucket, R-forge, and Github
- You can install a package using command; or use the "packages" tab on bottom right panel
- To use/load a package, you can use comment; or click on the checkbox in the "packages" tab



DataAnalytics Package

- You need to install the "package" made specifically for this course: `DataAnalytics`. This package contains nearly all of the datasets needed for the course
  - **Acknowledgement**: This package is developed by former instructor of this course, Professor Peter Rossi
- To install the package, you will first need to install the package, `remotes`, from CRAN and then use a special command to load the package

DataAnalytics Package

- The easiest way to do this is to execute the first block of code in `Ch 1 code snippets.R`

```
##############################################
# code snippets for Chapter 1
##############################################

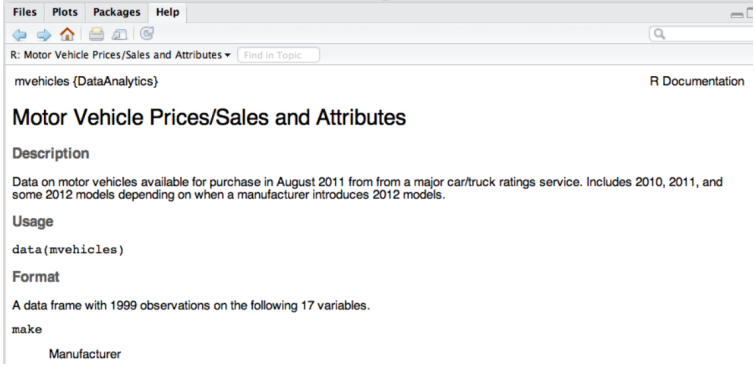
# ----- install course package ----------------------------------------------------------
install.packages("remotes")                          # install remotes package
library("remotes")                                   # "load" package
install_bitbucket("perossichi/DataAnalytics")        # install the course package
                                                     # note: this will take a few
                                                     # depending on internet conne


library("DataAnalytics")
```

- When you do this, you will see the commands being executed in the console window. You may get a warning but don't worry
- This may take 5-30 seconds depending on your internet connection speed
- **You only need to install a package once!**

DataAnalytics Package

- To verify that it worked, check that you can access the `mvehicles` dataset, by typing `?mvehicles` into the console
- The help file should appear in the help window. The help file tells you that this is data on about cars and trucks

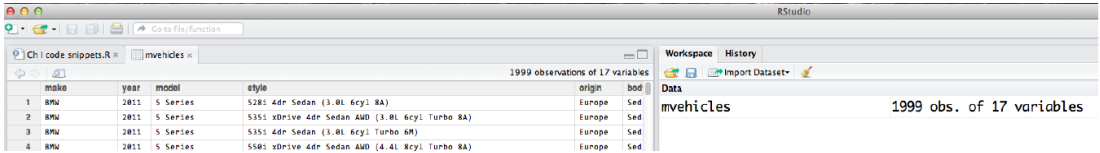


The R workspace

- In R, all data and code should be loaded into memory (the "workspace") and then executed
- RStudio displays what is "in" the workspace in the upper right corner
- All datasets must be "loaded"
  - If a dataset is already in `DataAnalytics`: Use the `data()` command to load
  - If a dataset is in other format (e.g., CSV): Use the `read.csv()` command to load

The R workspace: An Example

- Let's load `mvehicles` and see that it is in the workspace. Here it is:



- I used the command `data(mvehicles)` to load it and I simply double-clicked "mvehicles" in the "Workspace" tab to display it's contents
- It looks like `mvehicles` is an array (like a worksheet) with rows corresponding to the observations and columns corresponding to variables
- In R, data is stored in a special type of object called a "data frame"

Objects and Functional Orientation

- R implements a limited form of object-oriented programming
- Everything in R is an "object". This includes both data and programs
- To do something in R means to apply a **function** to an **object**

$$Object_1 = \text{function}(Object_0).$$

Common Data Objects in R

- **Scalars**: a single value (just a vector of length 1)
- **Vectors**: one-dimensional collection of values
- **Matrices**: two-dimensional array; all rows (columns) have the same number of values
- **Data Frames**: table-like structure with rows and columns

```
# x1 and x2 are scalars
x1 = 3; x2 = 4
# Combine scalars into vectors y1 and y2
y1 = c(x1,x2); y2 = c(x2,x1)
# Combine vectors into a matrix z
z = cbind(y1,y2)
# Example of data frame: mvehicle from DataAnalytics (see more in example function slides)
```

Functions

- You may have noticed that all of the R commands use functional notation, e.g `data(mvehicles)` like f(x). `data` is the function and `mvehicles` is the input
- Executing a function is termed "calling the function". In programming, some functions are called for their effects and not their output. `data()` is one such function. All `data()` does is load the dataset into the workspace, making its contents available

Function Examples: `head()`

- `head()`: this function prints the "head" (by default first six elements) of any R object

```
head(mvehicles)
##   make year    model                                          style  origin bodytype      emv seats roo
## 1  BMW 2011 5 Series                   528i 4dr Sedan (3.0L 6cyl 8A)  Europe    Sedan 45362.90     5 0.3
## 2  BMW 2011 5 Series 535i xDrive 4dr Sedan AWD (3.0L 6cyl Turbo 8A)  Europe    Sedan 54518.63     5 0.3
## 3  BMW 2011 5 Series              535i 4dr Sedan (3.0L 6cyl Turbo 6M)  Europe    Sedan 50066.94     5 0.3
## 4  BMW 2011 5 Series 550i xDrive 4dr Sedan AWD (4.4L 8cyl Turbo 8A)  Europe    Sedan 61599.65     5 0.3
## 5  BMW 2011 5 Series              550i 4dr Sedan (4.4L 8cyl Turbo 6M)  Europe    Sedan 60560.78     5 0.3
## 6 Ford 2011     Edge                   Limited 4dr SUV (3.5L 6cyl 6A) America      SUV 35441.33     5 0.3
out = head(mvehicles)
```

Function Examples: `rnorm()` and `summary()`

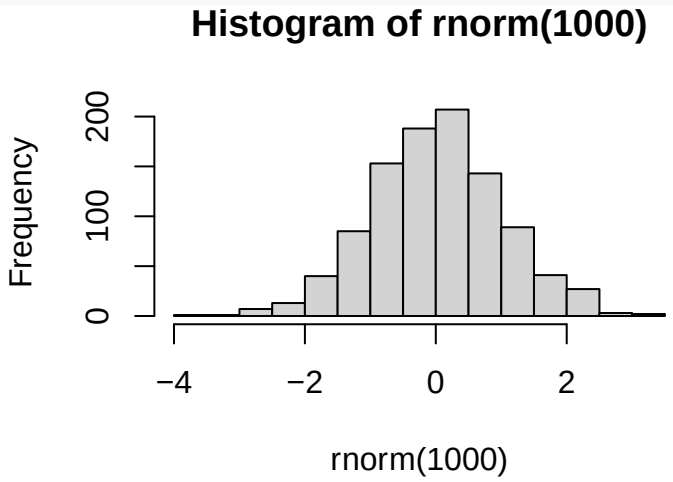
- `rnorm(n)`: generates n $N(0, 1)$ random numbers
- `summary()`: provides summaries of the input (can be data, regresion results, etc.)
- Use `$` to refer to a specific variable within a data frame: e.g., `mvehicles$sales`

```
rnums = rnorm(1000)
summary(rnorm(10))
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -2.3463 -0.6166  0.6105  0.1204  1.0007  1.3334
summary(mvehicles$sales)
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       6     288    1166    3730    3606  152232
```

Function Examples: `hist()`

- `hist()`: compute a histogram of the given data value

```
hist(rnorm(1000)) #  plot 1000 normal numbers
```



Appendix

Glossary of R Commands

- `c(1,2,3)` make or "concatenate" a vector with the objects 1, 2, 3. note: the objects must be of the same type.
- `data(A)` allows access to data frame A that is in the data library.
- `head(A)` returns the first parts of a data frame A.
- `function(args){}` define a function of the specified args (there can be more than one (separate by commas) using the statements in between the braces.
- `hist(variable,breaks=xx,xlab="",ylab="",main="")` histogram of variable with approx breaks number of bins, labels for axes and main title
- `library("DataAnalytics")` loads the R package containing datasets and customized functions for our class.
- `rnorm()` generates pseudo normal random numbers.

Glossary of R Commands

- `str(object)` tells you the "structure" of an object, e.g. str(dataframe).
- `sum(x)` computes the sum of elements in x.
- `var(X), mean(X), sd(X)` Compute the variance, mean, standard deviation of a variable named X.